



VERGE in VBS 2026

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Abstract. This paper describes VERGE, an interactive system for retrieving video content, supporting searches over images extracted from videos. The system retains its core retrieval modalities and fusion techniques in an improved form, while introducing new modalities, including optical character recognition, video question answering, underwater object detection, image quality-based retrieval, and surgical scenes understanding. VERGE's web application continues to offer its user-friendly interface, helping users create queries and efficiently explore the top search results. The system now incorporates newly extracted keyframes for the V3C dataset, generated with a new algorithm developed for this purpose.

Keywords: video search engine · video retrieval · text to video

1 Introduction

VERGE is an interactive video retrieval system that allows users to search images extracted from videos through a web application. It supports creating queries, exploring top-ranked results, and submitting the desired content. The system has participated in the Video Browser Showdown (VBS) competition [20] since its start in 2012 and, in 2024, also joined the IVR4B special session at the CBMI conference¹, where it was awarded first place. In the current version, the focus was on improving the system's performance in several ways. First, new keyframes were extracted for the V3C dataset using a novel algorithm developed specifically for this purpose. Second, most of the existing retrieval modalities were enhanced and new ones were introduced, mainly for the Marine Video and LapGynLHE datasets. Finally, the user interface was upgraded to support all new retrieval methods.

¹ https://cbmi2024.org/?page_id=100#IVR4B.

The structure of the paper is as follows. Section 2 introduces the VERGE framework, while Sect. 3 focuses on keyframe extraction for the V3C dataset. Section 4 describes the retrieval modalities and Sect. 5 briefly presents the user interface and its features. Finally, Sect. 6 discusses future work.

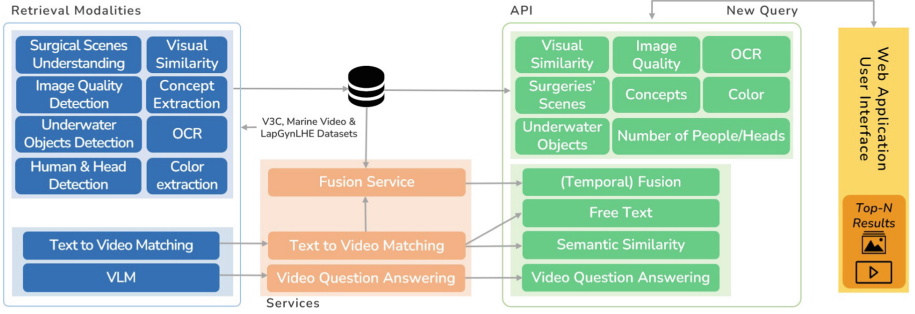


Fig. 1. The VERGE framework

2 The VERGE Framework

VERGE follows the same four-layer architecture as in previous years, shown in Fig. 1. The first layer consists of the retrieval modalities that analyze the data, with most of them storing the results in a database. The second layer contains the database together with the services, which include text-to-video matching, video question answering, and fusion. The third layer is the API, which connects these components with the fourth layer, the web application that enables users to formulate queries and view ranked results. VERGE supports all the available VBS datasets, including V3C [21] (from the Vimeo Creative Commons Collection), Marine Video [25] (underwater/scuba diving), and LapGynLHE [15] (surgeries in laparoscopic gynecology).

3 Keyframe Extraction for V3C

This module implements an end-to-end keyframe extraction pipeline that outputs a compact set of perceptually, and semantically diverse keyframes. The process begins with a hierarchical temporal segmentation of the video: shots are detected using a learned shot-boundary detector (TransNetV2 [23]), subshots are derived from DCT-based features, and Very Similar Segments (VSS) are identified within each subshot. For each VSS, the sharpest frame is chosen as its provisional keyframe. Deep CNN embeddings from a pretrained ResNet152 [7] are then extracted for all provisional VSS keyframes. These candidate frames are added to the final collection of representative keyframes only if their Euclidean

distance to the already selected keyframes in the embedding space exceeds a predefined threshold and their sharpness, measured via the variance of Laplacian, surpasses a minimum quality requirement. In addition, simple but effective heuristics, such as temporal spacing restrictions and the exclusion of shot-transition frames, are applied to further refine the selection of keyframes.

4 Retrieval Modalities

Some modalities from previous years remain unchanged, while others have been improved, and several new ones were introduced. The unchanged modalities are only briefly described here (see [17] for details). The new modalities include Video Question Answering (VQA), Optical Character Recognition (OCR), Underwater Objects Detection, Image Quality-Based Retrieval, and Surgical Scenes Understanding. Not all modalities are applied across all datasets. The Surgical Scenes Understanding modalities are specific to the LapGynLHE dataset. The Image Quality-Based Retrieval and Underwater Objects Detection modalities are specific to the Marine Video dataset, which also supports Text to Video matching, Concepts, Color-Based search, Semantic Similarity, and OCR. The V3C dataset supports all modalities except those that are dataset-specific. Across all datasets, Visual Similarity and Fusion are available.

4.1 Unchanged Retrieval Modalities

Visual Similarity Search. This module uses features from the final pooling layer of GoogleNet (DCNN) and employs IVFADC [10,18] to retrieve similar images.

Semantic Similarity Search. In this type of search, a video keyframe is used as the query, and the system retrieves semantically related keyframes. The retrieval is based on cross-modal embeddings produced by the Text to Video Matching Module, enabling comparisons that capture not only visual similarity but also deeper semantic relationships.

Late Fusion Approaches for Modalities. Late Fusion combines multiple modalities, including text, concepts, color, and faces, within or across keyframes to improve search [17].

4.2 Improved and New Retrieval Modalities

Text to Video. This module processes free-text queries to retrieve relevant video shots in two stages: an offline video indexing stage and an online query processing stage. This year, we replaced the complex trainable module with a simpler yet more effective architecture, and leveraged the capabilities of a Video LLM. The workflow begins with an offline video indexing step where five families of vision-language models (VLMs) with pre-trained models are employed to

extract video shot embeddings. In the online query processing step, the input query is processed by a Retrieval Module (RM) ([6]), which leverages the same VLMs to encode textual embeddings. For each model family, a Similarity Module (SM) computes similarity scores between the query embeddings and the stored visual ones and aggregates them into a single value per family. The resulting five scores are fused via a Learnable Weighting Network (LWN) [6], yielding a unified similarity score. Then the retrieved video shots are refined using Qwen2.5-VL [1], a large multimodal model capable of evaluating fine-grained semantic alignment. Qwen2.5-VL produces contextual relevance scores, and the list is reordered accordingly.

Human and Head Detection. The module operates with the ultimate goal of identifying and counting the number of both humans and human heads in each keyframe, thereby differentiating between single and multiple humans within keyframes. Compared to last year’s module, the YOLOv8 [27] model, fine-tuned with the CrowdHuman [22] dataset, was replaced with the latest YOLOv12 [24] architecture, pretrained on MS COCO [14] for human silhouette detection. For human heads, a larger-scale and diverse dataset was generated by combining existing databases from public sources used for fine-tuning a second YOLOv12 model, which is specialized in this specific task.

Concept-Based Retrieval. Last year’s ensemble of specialized concept detectors was replaced with a single Qwen2.5-VL model [1], specifically the 7B parameters variant. A compact curated set of about 50 target concepts relevant to the task was compiled. For each keyframe, Qwen2.5-VL was queried with a short natural language prompt for each concept. The model’s textual response was parsed to produce a single per keyframe confidence score for that concept, which was used instead of the probabilities previously calculated by the dedicated detectors.

Color Extraction. This module splits each image into 9 tiles and extracts features from each subdivision using CLD and Lab colors [9], which better reflect human perception than RGB [3]. It computes dominant color families per tile, compares images to reference centroids, and outputs per-image summaries. Adding Lab features improves accuracy over the previous year’s CLD-only model.

Optical Character Recognition. The OCR module detects and extracts the textual content visually depicted in videos, enabling filtering capabilities for differentiating between scenes and providing a more detailed understanding of context-driven activities. PaddleOCR [5] is a highly efficient, multilingual, and versatile framework that performs text detection and recognition in a single pipeline, achieving a high level of robustness against variations in fonts, orientations, and backgrounds.

Video Question Answering. The VQA module is responsible for providing answers to open-ended queries about the contents of a video. Given a video that is retrieved and the user believes it is related to the textual query, VQA processes the entire video and generates an answer. To this end, Qwen3-VL 8B Instruct [28], an instruction tuned Video LLM, capable of perceiving and understanding long videos, was leveraged. In addition, the Video LLM was instructed to provide the timestamps of the relevant video segments, in order to temporally ground the responses. This modality will be very useful for the Visual Question Answering task type.

Underwater Objects Detection. The module is introduced to detect the presence of common underwater objects (e.g., fish, sharks, corals, and divers) in each keyframe of the Marine Video dataset to enable users to filter images based on whether they contain (or not) objects, their classes, and occurrences. The YOLOv12 architecture was deployed [24], fine-tuned on a publicly available underwater dataset [12] suitable for object detection tasks. It also integrates a fish species detection submodule, enabling the retrieval of keyframes depicting specific fish species.

Image Quality-Based Retrieval. This module ranks each keyframe of the Marine Video dataset according to the intensity of the blur that occurs as a result of the forward scattering effect. It allows the users to distinguish between sharp and blurred underwater images, considering the underwater image sharpness measure (UISM) [16]. UISM assigns lower values (closer to zero) to heavily blurred images and higher values to sharper images that are less affected by underwater forward scattering.

Surgical Scenes Understanding. This module detects surgical instruments in the LapGynLHE dataset, extracts surgical phases and actions, and detects various organs, providing users with efficient indexing and retrieval-based capabilities. For the detection and localization of surgical instruments, YOLOv12 was fine-tuned on a custom dataset [19] constructed by combining publicly available datasets [11, 26]. A similar approach was employed for the detection of organs, using the Cholec 8K Segmentation [8] dataset. Moreover, a ResNet50 [7] architecture, fine-tuned on a publicly available dataset [13], was employed to classify the keyframes according to the total number of depicted instruments, as well as foundation models [2, 4] to recognize surgical phases and actions.

5 User Interface

The VERGE Web application maintains the same structure as in our previous work [17] and is shown in Fig. 2. It is divided into two main parts: the menu on the left and the results panel on the right. The system provides both a novice and an advanced mode. At the top of the menu are the logo, a settings button

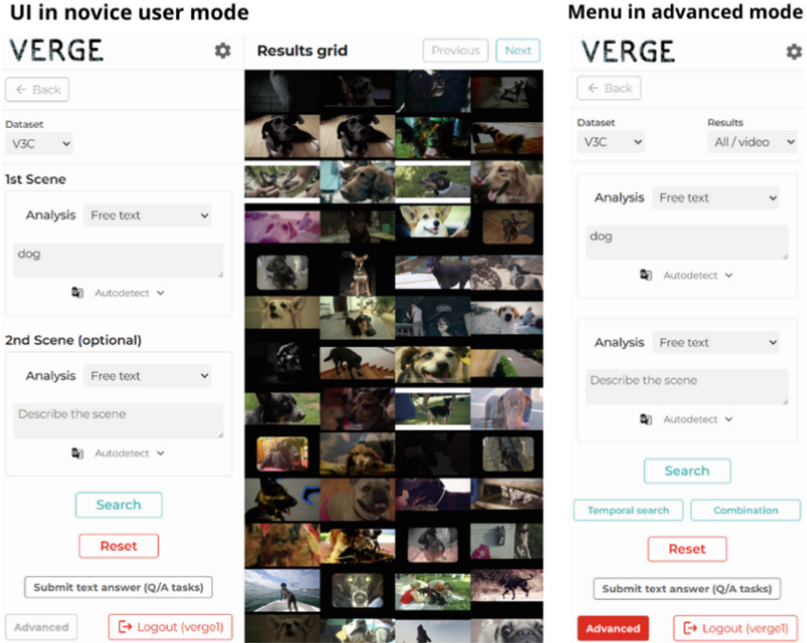


Fig. 2. The VERGE User Interface

to adjust the zoom level of the results, a history back button, and the dataset selector. The main part of the menu consists of two query containers for forming queries. Each container includes a dropdown menu with the available analyses and a corresponding input field for the selected analysis. Below the containers are the ‘Search’ and ‘Reset’ buttons: ‘Search’ performs a temporal search if both containers have input, otherwise it runs a standard search, while ‘Reset’ returns the containers to their default state. At the bottom of the menu are the ‘Submit text answer’ and ‘Advanced’ buttons. The former allows users to submit textual answers, while the latter switches the application to advanced mode. In advanced mode, the results format selector becomes available, enabling users to choose between retrieving all matching images or only the first per video. Additional analyses also become available, and users can choose to perform either a temporal search or combine two queries using the ‘Temporal’ and ‘Combination’ search buttons respectively. The results panel on the right displays the retrieved results and allows users to browse them.

6 Future Work

In the coming months, our work will focus on applying the analyses to the new V3C keyframes, while testing and improving the new modalities, and integrating them into the web application.

Acknowledgments. This work is supported by the Horizon Europe program under the grant agreements 101096405 UP2030, 101182365 NAMOR, 101181380 AQUAMON, 101070190 AI4Trust, 101070109 TransMIXR, and 101073952 PERIVALLON.

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